

## Taylor Ertrachter

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## Technical Skills

- **Languages:** C, C++, Rust, Bash, Python 3, JavaScript, PHP, Java, MySQL, QT (C++, QML)
- **Technologies:** Embedded Linux (ARM), Linaro ARM Compiler, Docker, Canonical LXD, Git, Atlassian (Jira, Confluence), MariaDB, Jenkins, Apache
- **AWS Services:** EC2, S3, CloudWatch, Route53, Pinpoint, Polly, Backup, Billing and Cost Management, IAM
- **Hardware Interfaces:** I2C, SPI, GPIO, JTAG, Logic Analyzers

## Education

Florida Institute of Technology | 2017 - 2021

Bachelor's degree, Computer Software Engineering

Minor in Business Management

## Professional Experience

Data Flow Systems, Inc. – a Badger Meter Brand

Product Engineering Manager | July 2022 – Present

- Leads a team of seven engineers across hardware, software, and testing subjects to deliver on product goals, performing annual **SMART goals** to maximize individual success.
- Collaborates with Product Managers to perform **SCRUM** activities including designing sprints and quarterly department goals for the team using **Jira** and **Confluence**.
- Recruited, onboarded, and coached new employees, successfully integrating them into the team.
- Provides technical and strategic direction for the software development of smart Remote Terminal Unit products (RTU) that provide **SCADA telemetry** over proprietary radio, cellular, and LAN based communications.
- Maintained principal RTU product backend using **ARM Embedded Linux OS** with PHP, C, C++ programs using **Linaro ARM GNU toolchain**. This product also includes a touchscreen interface GUI made with **QT**.
- Developed a critical firmware update deployed using **Debian packages** that allowed the principal RTU to interface with **VFD pump controllers**, enabling new market opportunities.
- Created **Docker images** for both the RTU development and testing environments, allowing for consistent compilation and the emulation of products without physical hardware. These images

were integrated with the **Development Containers open specification** to define reproducible development environments in **VSCode**.

#### **Software Engineer | January 2022 – July 2022**

- Led a team in migrating a legacy **Debian-based SCADA** server system to a modern, containerized architecture using **Canonical LXD** hosted on **Amazon Web Services**.
- Developed a migration strategy that optimized system performance, scalability, and security using AWS services like **EC2**, **S3**, **Backup**, and **Route53** while allowing existing customer data to be preserved during migration.
- Implemented proactive monitoring and alerting mechanisms to identify and resolve system downtime using **Cloudwatch**.
- Created a comprehensive customer billing structure using **Billing and Cost Management**.
- Provided comprehensive documentation and training to internal teams for the upgraded, containerized system using **Confluence**.

#### **Associate Software Engineer | May 2021 – January 2022**

- Designed and upgraded the back-end of a patented legacy product allowing autonomous RTU operation using a **MariaDB database** and a **RESTful API** built in **Rust** to communicate with a SCADA server front-end running on **Node.js**.
- Collaborated with internal teams to define product upgrade requirements and ensure alignment with customer needs.
- Worked with testing teams to verify functionality using an in-house **Python** simulation program.
- Developed comprehensive product documentation, including technical specifications and API guides, to facilitate adoption and support front-end integration using **rustdoc**.

#### **Software Developer Intern | November 2020 – May 2021**

- Designed and developed an automated regression testing program in **Rust**, streamlining the **Excel** based testing process for an existing product.
- Upgraded a legacy landline modem-based SCADA alert system to utilize **Voice over Internet Protocol (VoIP)** using **Twilio** and **Python 3**, modernizing a key product component by allowing dynamic speech-based alarm messages to be created live using **AWS Polly** and text-based SMS or email alarm messages using **AWS Pinpoint**.

### **Additional Experience**

- **Gleason Performing Arts Center Technician** | Florida Institute of Technology (*June 2019 - October 2020*)
- **Programmer Analyst and Help Desk Technician** | Stellar (*May 2018 - July 2018*)

- **Tech Support Center Analyst** | Florida Institute of Technology (*August 2017 - July 2019*)

## **Volunteer Experience**

### **Boy Scouts of America**

- **Nova Counselor and Supernova Mentor** | Central Florida and Greater Alaska Council (*August 2021 – January 2025*)
- **Merit Badge Counselor** | Central Florida and Greater Alaska Council (*August 2021 – January 2025*)
- **Assistant Scout Master** | North Florida Council, Troop 37 (*Present*)